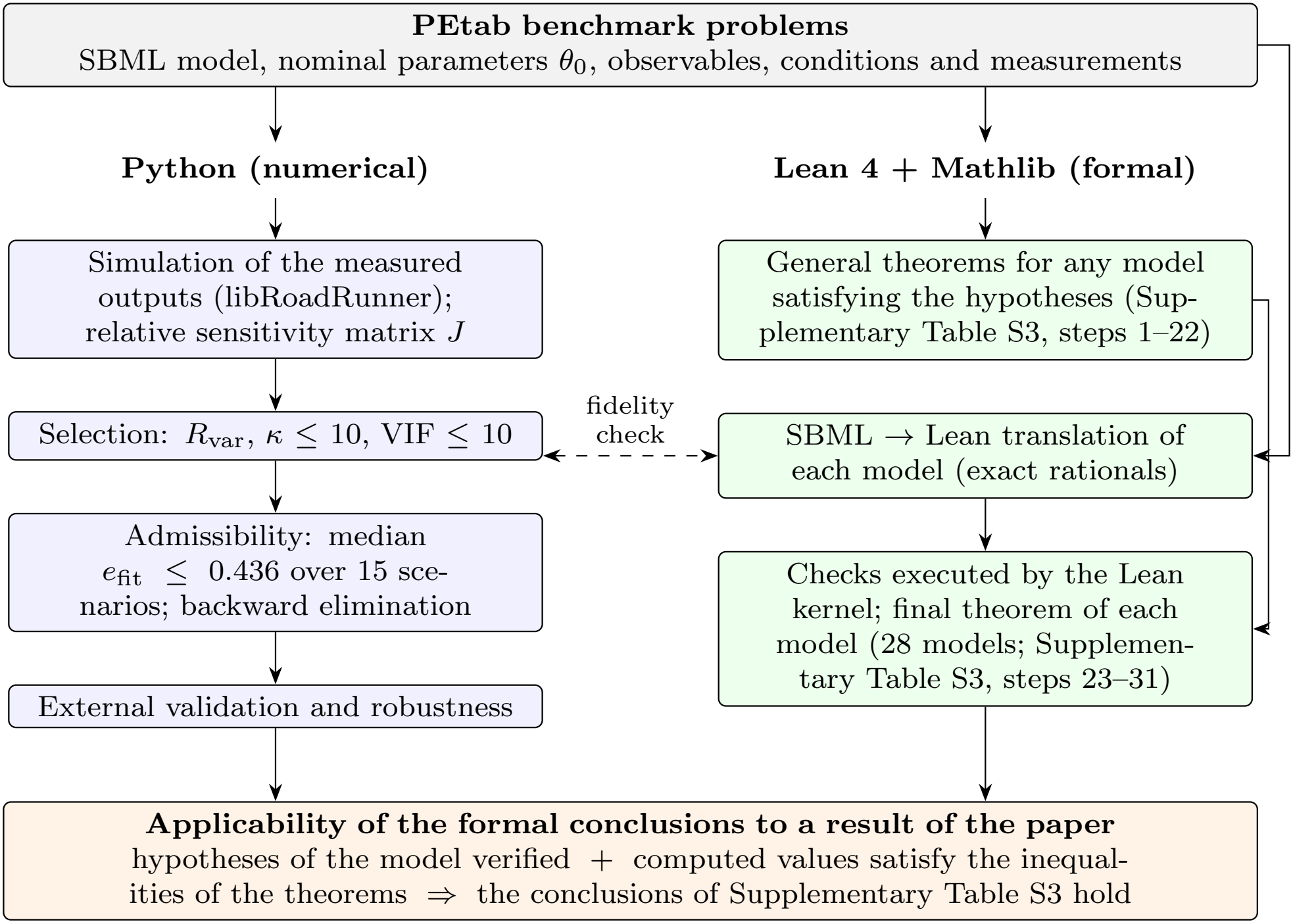

PEtab benchmark problems
SBML model, nominal parameters θ_0, observables, conditions and measurements
Python (numerical)
Lean 4 + Mathlib (formal)
Simulation of the measured outputs (libRoadRunner); relative sensitivity matrix J
General theorems for any model satisfying the hypotheses (Supplementary Table S3, steps 1–22)
Selection: R_{var}, $\kappa \leq 10$, VIF ≤ 10
fidelity check
SBML → Lean translation of each model (exact rationals)
Admissibility: median $e_{\mathrm{fit}} \leq 0.436$ over 15 scenarios; backward elimination
Checks executed by the Lean kernel; final theorem of each model (28 models; Supplementary Table S3, steps 23–31)
External validation and robustness
Applicability of the formal conclusions to a result of the paper
hypotheses of the model verified + computed values satisfy the inequalities of the theorems ⇒ the conclusions of Supplementary Table S3 hold